

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Applied Physics

Examination: 1st Periodical Examination

Branch: CSE/ECE/CE

Course Code: PHY-111

Scheme: NEP/CBCS

Semester: 1st

Question Paper for 1st Periodical Examination (October 2025) (Session: Aug.-Dec. 2025)

Time: One and a half hours

Max Marks: 30

Note: Attempt all questions

CO	Course Outcomes
CO1	Recognize the importance of Physics in describing different physical phenomenon
CO2	Comprehend the Special theory of relativity by examining inertial & non-inertial frames, Michelson-Morley experiment, Galilean & Lorentz transformations.
CO3	Demonstrate the concept of harmonic oscillators by studying simple harmonic, damped harmonic and forced harmonic oscillations.
CO4	Illustrate the concept of LASER, Optical Fibres and X-rays in various fields.
CO5	Appraise the concept of electromagnetic waves and Superconductivity in the field of Engineering
CO6	Recognize the inadequacy of classical mechanics for certain physical problems and thus find the solution of these problems using Quantum Mechanics

Que No.	Questions	Marks	Target CO	Target KL
1.(a)	Define sharpness of resonance in context to forced oscillator.	1	CO3	L1
1.(b)	What will be the effect on the time period of simple pendulum if mass is doubled.	1	CO3	L2
1.(c)	Discuss logarithmic decrement.	1	CO3	L2
1.(d)	Write the equation representing a simple harmonic progressive wave and explain various symbols appearing in it.	1	CO3	L1
1.(e)	Find the maximum value of the velocity & acceleration of a particle executing SHM of time period of 10π s and of amplitude 1cm.	1	CO3	L2
2.	Derive an expression for the total energy of harmonic oscillator and show that it is constant and proportional to square of the amplitude.	5	CO3	L2
3.	What are main components of any laser system? With the help of schematic diagram discuss the role of each component in working of laser.	5	CO4	L2
4.	What are Einstein coefficients? Derive the relation between Einstein's coefficients.	5	CO4	L3
5.	Explain in detail the principle, construction, working & output of Ruby Laser. Or Distinguish between the damped and forced oscillations. Derive differential equation of forced Oscillation and find its solution.	10	CO4 / CO3	L2

Course Advisor: Dr. Neelam Guleria

Designation: AP (AS&H) Physics

CO: Course Outcome

KL: Knowledge Level: L1 – Remember L2 – Understand L3 – Apply L4 Analyze L5 – Evaluate L6 – Create